

Hippocrates Insurance+

Course: Life Insurance Mathematics Practice

Luisa Scaroni

Chao Gao

Maria Theodosiou

Kit Ching Choy

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Abstract

Abstract: This project presents the design and actuarial valuation of **Hippocrates Insurance+**, a hybrid endowment policy combining guaranteed capital (Branch 21) with investment-linked growth (Branch 23). The product uniquely integrates a wellness incentive program designed to reward policyholders for maintaining healthy lifestyle habits.

The actuarial pricing models are developed based on a mixed mortality table (XK) and a technical interest rate of 3.75%, employing the Equivalence Principle to derive Net Single Premiums, Periodic Premiums, and Commercial Tariffs with specific expense loadings.

Furthermore, the report analyzes the evolution of policy values (mathematical reserves) over a 20-year term and conducts sensitivity analyses regarding interest rate fluctuations (increasing to 4.75%) and gender-specific mortality risks. Finally, the study demonstrates a profit-sharing mechanism and evaluates critical business challenges, specifically focusing on mortality risk and interest rate risk.

Keywords: Endowment Insurance, Branch 21/23, Actuarial Pricing, Policy Value, Sensitivity Analysis, Profit Sharing.

1 PART 1

1.1 Product Overview

Hippocrates Insurance+ is an endowment life insurance policy that combines **Branch 21** (guaranteed capital) and **Branch 23** (investment-linked) features.

This product offers lump-sum (capital) or annuity options at death or at maturity. A distinctive feature is that it rewards sustained physical and mental wellness habits with a transparent, privacy-respecting bonus

1.2 Target Client Profile

This product is designed for clients who seek:

- **Family Protection:** Reliable life coverage to safeguard loved ones in case of premature death.
- **Disciplined Wealth Building:** Valuing structured, long-term saving with capital guarantees to ensure financial security.
- **Inflation-Resilient Growth:** Desiring exposure to market-linked returns to preserve and grow wealth over time.
- **Wellness-Driven Motivation:** Open to incentives that reward healthy lifestyle choices—to be healthy both physically and mentally.

1.3 Contractual Parties

The contract involves the following parties:

1. **Insurer:** Hippocrates Life & Wellness Insurance.
2. **Policyholder:** The person who signs the contract between him/her and the insurance company. He/she is the one who will be paying the premium. And if the insured is still alive at the end of the contract term, He/she receives benefits.
3. **Insured:** The person whose life or death determines the contract. Note: The Insured can be different from the Policyholder.
4. **Beneficiaries:** The recipients of the policy proceeds when insured dies during the term of the contract.

1.4 Fund Allocation Strategy

The premium is allocated to balance security and growth potential:

- **70% Branch 21:** Guaranteed capital & possible annual profit sharing.
- **30% Branch 23:** Investment-linked funds with a 0% minimum return guarantee, offering growth potential.

1.5 Premium Structure

- **Payment Frequency:** Options include a Single lump sum payment or regular payments over 10 years.
- **Grace Periods:** For regular payments, grace periods are available for clients.
- **Opt-in Wellness Bonus:**
 - Clients can connect to an optional health & mental wellness app (tracking steps, sleep, mood check-ins).
 - **Incentive:** If yearly goals are met (e.g., 8,000 steps/day average, healthy BMI range, regular resilience check-ins), policyholders can receive premium discounts.

1.6 Benefits and Maturity

The policy term is set to **20 years** (Maturity). The payout structure is defined as follows:

1.6.1 Death Benefit (Death before Maturity)

If the insured dies during the policy term:

- The beneficiary receives a lump sum payout consisting of the Branch 21 guaranteed capital plus the Branch 23 account value.
- **Minimum Guarantee:** There is a minimum death benefit of at least the Branch 23 premiums paid.
- Payment is made to the beneficiary at the end of the year of death.

1.6.2 Survival Benefit (Alive at Maturity)

If the insured survives to the end of the term:

- **Option A:** Lump sum payment of Branch 21 guaranteed capital + Branch 23 account value to the policyholder.
- **Option B:** Lifetime annuity.

1.7 Illustrative Scenario: The Case of Paul

Background: Paul is a 35-year-old father of two and the family's main breadwinner. He worries about not being there to support his children as they grow up.

Solution: Hippocrates Insurance+ gives him reliable life protection while building disciplined savings, ensuring his loved ones are financially secure whether he lives to maturity or passes away prematurely.

Contract Arrangement:

- **Policyholder & Insured:** Paul
- **Beneficiaries:** Paul's wife and 2 children

2 PART 2

2.1 Actuarial Basis and Assumptions

The pricing model relies on the following assumptions:

- **Mortality Table:** XK(MK and FK mixture).

We assign weights of 0.65 and 0.35 to men's and women's survival probability, respectively. (Here, we didn't use the weights on the MK and FK's $l(x)$, because this method does not directly account for age-specific risks by gender, potentially underestimating or overestimating mortality rates for certain age groups.)

We can calculate:

$$\begin{aligned}
 {}_nE_x(XK) &= v^n {}_nP_x \\
 &= v^n (0.65 \cdot {}_nP_x(\text{Male}) + 0.35 \cdot {}_nP_x(\text{Female})) \\
 &= v^n 0.65 \cdot {}_nP_x(\text{Male}) + 0.35 \cdot {}_nP_x(\text{Female}) \\
 &= 0.65 \cdot {}_nE_x(MK) + 0.35 \cdot {}_nE_x(FK)
 \end{aligned}$$

Similarly, we can get

$$\begin{aligned}
 A_{x:\overline{n}|}^1(XK) &= 0.65 \cdot A_{x:\overline{n}|}^1(MK) + 0.35 \cdot A_{x:\overline{n}|}^1(FK) \\
 \ddot{a}_{x:\overline{n}|}(XK) &= 0.65 \cdot \ddot{a}_{x:\overline{n}|}(MK) + 0.35 \cdot \ddot{a}_{x:\overline{n}|}(FK)
 \end{aligned}$$

- **Beneficial Payment time:** We will be using XK(K+1), meaning compensation will be paid at the end of the year of death.
- **Technical Interest Rate (i):** Assumed TWO situations, one is to be 3.75% per annum, the other is to be 4.75% per annum.
- **Discount Factor (v):** $v = \frac{1}{1+i}$.

- **Fee structure:** Type "death"
- **In case of death before the end date:** $KO = KL$

2.2 Mathematical Model

2.2.1 Notation

Let:

- x : Age at entry.
- n : Policy duration in years.
- m : Payment duration in years.
- UP : Single Premium.
- PP : Periodic Premium.
- Inventory Charge:
 - 1) b_1 : On the reserve, (decrease the technical interest rate)
 - 2) b_2 : On the death capital
- f : flat rate
- c : Commission Fee
- Capital Life: $KL = KO$
- Dotation = $d\%$ on the policy value (Res)

2.2.2 Derivation

Assuming an **Endowment Assurance** contract with net single premium:

1. Inventory Premiums:

$$P' = KL \cdot ({}_nE_x(i^*) + A_{x:\overline{n}|}^1(i^*) + b_2 \cdot \ddot{a}_{x:\overline{n}|}(i^*)) \quad (1)$$

where, $i^* = i - b_1$

2. Commercial Premium: This consists of acquisition charge:

$$P'' = \frac{P'}{1 - c} \quad (2)$$

3. Single Premium:

$$UP = P'' + f \quad (3)$$

If the contract with periodic premium,

4. Periodic Premium:

$$PP = \frac{P'' + m \cdot f}{\ddot{a}_{x:\overline{m}|}} \quad (4)$$

2.3 Numerical Illustrations

Here I'll give an example at age 35; the rest are calculated in a similar manner.

2.3.1 Parameters

We calculate the premium for a standard policyholder:

- $x = 35$ years
- $n = 20$ years
- $m = 10$ years
- $KL = 25,000$
- $b_1 = 0.10\%$
- $b_2 = 0.001$
- $f = 2.5$ EUR
- $c = 0.13$
- $i = 3.75$

2.3.2 Calculation Results(Age 35 - one example)

Based on the assumptions, the factors are calculated as follows (by Excel Calculation on Toledo):

$$\begin{aligned} \ddot{a}_{35:\overline{20}|}(XK_{i^*=3.65}) &= 0.65 \cdot \ddot{a}_{35:\overline{20}|}(MK_{i^*=3.65}) + 0.35 \cdot \ddot{a}_{35:\overline{20}|}(FK_{i^*=3.65}) \\ &\approx 0.65 * 14.297555 + 0.35 * 14.090866 \\ &\approx 14.16321 \end{aligned}$$

$$\begin{aligned}
\ddot{a}_{35:\overline{10}|}(XK_{i^*=3.65}) &= 0.65 \cdot \ddot{a}_{35:\overline{10}|}(MK_{i^*=3.65}) + 0.35 \cdot \ddot{a}_{35:\overline{10}|}(FK_{i^*=3.65}) \\
&\approx 0.65 * 8.500410 + 0.35 * 8.453684 \\
&\approx 8.470038
\end{aligned}$$

$$\begin{aligned}
A^1_{35:\overline{20}|}(XK_{i^*=3.65}) &= 0.65 \cdot A^1_{35:\overline{20}|}(MK_{i^*=3.65}) + 0.35 \cdot A^1_{35:\overline{20}|}(FK_{i^*=3.65}) \\
&\approx 0.65 * 0.034649 + 0.35 * 0.064914 \\
&\approx 0.05432125
\end{aligned}$$

$$\begin{aligned}
{}_{20}E_{35}(XK_{i^*=3.65}) &= 0.65 {}_{20}E_{35}(MK_{i^*=3.65}) + 0.35 {}_{20}E_{35}(FK_{i^*=3.65}) \\
&\approx 0.65 * 0.461867 + 0.35 * 0.438881 \\
&\approx 0.4469261
\end{aligned}$$

The resulting Single Tariff UP is:

$$\begin{aligned}
UP &= \frac{P'}{1-c} + f \\
&= \frac{KL \cdot ({}_nE_x(i^*) + A^1_{x:\overline{n}|}(i^*) + b_2 \cdot \ddot{a}_{x:\overline{n}|}(i^*))}{1-c} + f \\
&\approx \frac{25,000 \cdot (0.4469261 + 0.05432125 + 0.001 \cdot 14.16321)}{1-0.13} + 2.5 \\
&\approx 14813.15 \text{ EUR}
\end{aligned}$$

The resulting Periodic Tariff PP is:

$$PP = \frac{P'' + m \cdot f}{\ddot{a}_{x:m|}} = \frac{UP + (10-1) \cdot f}{\ddot{a}_{35:\overline{10}|}} \approx \frac{14813.15 + 9 \times 2.5}{8.470038} \approx 1751.54 \text{ EUR}$$

Other cases are calculated in a similar manner to the given example. For specifics, please refer to the R code.

2.4 Base Premium Calculations

In this section, we study the premium calculations for the Endowment Insurance based on our technical assumptions: a mixed mortality table (XK) and an interest rate of 3.75%,

considering a 20-year duration and a capital of 25,000 EUR.

The results are shown in Table 1.

Table 1: Base Premium Calculations (3.75%, Mixed Table)

Age	Single Premium (UP)	Periodic Premium (PP)	Total Cost ($10 \times PP$)	Extra Cost (Difference)
35	12,302.44	1,458.33	14,583.31	+2,280.86
45	12,702.05	1,522.76	15,227.58	+2,525.16
55	13,685.57	1,691.75	16,917.46	+3,231.93

First, we observe that premiums increase with age, demonstrating a positive correlation between the applicant's age and the premium amount ($UP_{55} > UP_{35}$). This phenomenon is driven by two key actuarial factors of Endowment policies:

- **Mortality Risk Dynamics:** q_x (the probability of death) increases exponentially with age because of the Makeham formula:

$$l_x = ks^x g^{c^x}$$

Indeed, the Term Insurance Component (${}_nA_x$) for a 55-year-old is higher compared to a 35-year-old, as the likelihood of a claim within the 20-year horizon is greater.

- **Present Value of Cash Flow:** Due to higher mortality, an older insured person implies an earlier expected payout. In addition, an earlier payout implies less time for the insurer to accrue interest on investments. To offset the loss of investment accumulation time, the upfront premium must be higher.

In general, the total cost of insurance when paid through periodic premiums is higher than the Single Premium (UP). This can be justified by two factors identified in our pricing model:

1. **Loss of interest:** With a single premium, the insurer receives the full capital upfront; on the other hand, with periodic payments, the capital is received gradually, generating less financial return over time.
2. **Mortality Risk during Payment:** If the policyholder passes away during the premium payment term, the obligation to pay future premiums ceases, but the insurer still must pay the death capital. For this reason, PP includes a loading to cover this specific risk.

2.5 Tariff Components Calculations

In this section, we calculate the Commercial Tariff Premiums, not merely the actuarial risk cost. The results are shown in Table 2.

Table 2: Commercial Tariff Premiums

Age	Inventory Premium	Commercial Premium	Single Tariff UP	Periodic Tariff PP
35	12,885.26	14,810.65	14,813.15	1,751.54
45	13,268.42	15,251.05	15,253.55	1,824.02
55	14,211.23	16,334.75	16,337.25	2,014.36

Analysing the results in Table 2, we attribute their values to financial discounting and expense loadings:

- **Impact of Acquisition Costs:** For every 100 EUR paid by the policyholder, 13 EUR covers operational distribution, meanwhile 87 EUR are allocated to cover the actuarial biometric risk and savings capital.
- **Mitigating Effect of Interest Rate:** The insurer collects less money today because they rely on investment returns to reach the 25,000 EUR capital at maturity.

2.5.1 Sensitivity Analysis

We performed a sensitivity analysis by modifying two variables: the interest rate and the mortality rate assumption. In the first case we chose an increased interest rate at a level of 4.75%; in the second, we considered the male mortality table (MK).

2.5.2 Increased Interest Rate (4.75%)

First, let's study the case with an interest rate of 4.75%. The results are shown in Table 3.

Table 3: Premium Calculations with Interest Rate 4.75%

Age	Single UP	Periodic PP	Total Cost	Extra Cost
35	10,389.24	1,281.39	12,813.89	+2,424.65
45	11,013.77	1,373.48	13,734.75	+2,720.89
55	12,488.27	1,604.78	16,047.80	+3,559.53

The results show a decrease of the premium following the increase in the interest rate, demonstrating an inverse relationship between the interest rate and the premium. Specifically, a 1% increase leads to a 15.5% reduction in the Unique Premium because of the financial discount effect: a higher interest rate implies that the insurer expects a greater return on investment, requiring also a lower initial capital contribution from the policyholder to guarantee the same final benefit of 25,000.

The impact on commercial tariffs is shown below in Table 4.

Table 4: Tariff Premiums with Interest Rate 4.75%

Age	Inventory Prem.	Commercial Prem.	Single Tariff UP	Periodic Tariff PP
35	10,763.47	12,371.81	12,374.31	1,523.02
45	11,191.82	12,864.16	12,866.66	1,601.10
55	12,248.52	14,078.76	14,081.26	1,805.45

The results obtained in Table 4 demonstrate that the reduction in the Net Premium (because of higher interest rate) leads to a decrease in the Inventory and Commercial Premiums. In addition, since c is calculated as a percentage of the premium, a lower base value means lower absolute costs for the policyholder.

2.5.3 Only Male Insured

Let's now consider a scenario where only men are insured. *Note: We acknowledge that since EU Gender Directive implementation insurance pricing must be strictly unisex, the following comparison is conducted only with the aim of comparing the different outcomes based on table.*

Table 5: Premium Calculations - Male Mortality Table (MK)

Age	Single UP	Periodic PP	Total Cost	Extra Cost
35	12,366.99	1,468.81	14,688.13	+2,321.14
45	12,841.61	1,546.21	15,462.13	+2,620.52
55	13,959.40	1,742.75	17,427.48	+3,468.08

The premiums calculated with the MK table are slightly higher compared to the mixed XK table. This outcome is expected due to the composition of the Endowment product:

- **Term Insurance Component:** Men have a higher mortality probability than women (and thus also than a mixed population). This increases the cost of the death coverage component.
- **Pure Endowment Component:** While higher probability reduces the cost of survival benefit, the increased cost of the death risk outweighs this reduction, resulting in a higher total premium.

Table 6: Tariff Premiums - Male Mortality Table (MK)

Age	Inventory Prem.	Commercial Prem.	Single Tariff UP	Periodic Tariff PP
35	12,947.15	14,881.78	14,884.28	1,763.35
45	13,402.20	15,404.82	15,407.32	1,850.44
55	14,473.61	16,636.33	16,638.83	2,071.93

As shown in Table 6, since the risk cost for men is higher, the Inventory Premium increases. Therefore, both Commercial and Tariff premiums rise. Furthermore, the increase in the biometric risk is amplified in the final tariff paid by the client because the expense loadings are applied as a percentage.

3 PART 3

Policy Value Calculation (Base Scenario) The policy value is given by the following formula:

$${}_tV = KL \cdot {}_{n-t}E_{x+t} + KO \cdot A^1_{x+t:n-t|} - PP \cdot \ddot{a}_{x+t:n-t|} \quad (5)$$

In this section, we calculate the policy value for the Endowment Insurance based on our technical assumptions: a mixed mortality table (XK) and an interest rate of 3.75%, considering a 20-year duration and a capital of 25,000 EUR. It is important to note that we have used the Net Periodic Premium in our calculations.

For a policyholder aged 35, we have calculated the following, using a weight of 0.65 for men and 0.35 for women to calculate the XK Policy Value.

The results are presented in Table 7.

Table 7: Policy Values over time ($i = 3.75\%$)

Time (t)	MK Policy Value	FK Policy Value	XK Policy Value
0	0.00	0.00	0.00
1	1,468.05	1,461.92	1,465.90
2	2,991.38	2,979.16	2,987.11
3	4,572.38	4,553.93	4,565.92
4	6,213.57	6,188.55	6,204.81
5	7,917.70	7,885.46	7,906.42
6	9,687.71	9,647.27	9,673.56
7	11,526.82	11,476.73	11,509.29
8	13,438.51	13,376.81	13,416.91
9	15,426.60	15,350.65	15,400.02
10	17,495.30	17,401.66	17,462.53
11	18,117.91	18,036.82	18,089.53
12	18,764.46	18,696.13	18,740.54
13	19,436.26	19,380.73	19,416.83
14	20,134.83	20,091.81	20,119.77
15	20,861.83	20,830.71	20,850.94
16	21,619.18	21,598.90	21,612.08
17	22,409.01	22,397.99	22,405.16
18	23,233.81	23,229.81	23,232.41
19	24,096.39	24,096.39	24,096.39
20	25,000.00	25,000.00	25,000.00

Observation: Comparing the 2nd and 3rd column, we easily notice that the policy

value for men is always higher. This is because men have a higher probability of dying. The last column gives us the policy value during the contract at each time $t = 0, 1, \dots, 20$.

3.1 Sensitivity Analysis (Interest Rate Risk)

3.1.1 Evolution of Policy Value

Below is the graph of the evolution of the policy value over the 20 years of the contract with an interest rate of 3.75%.

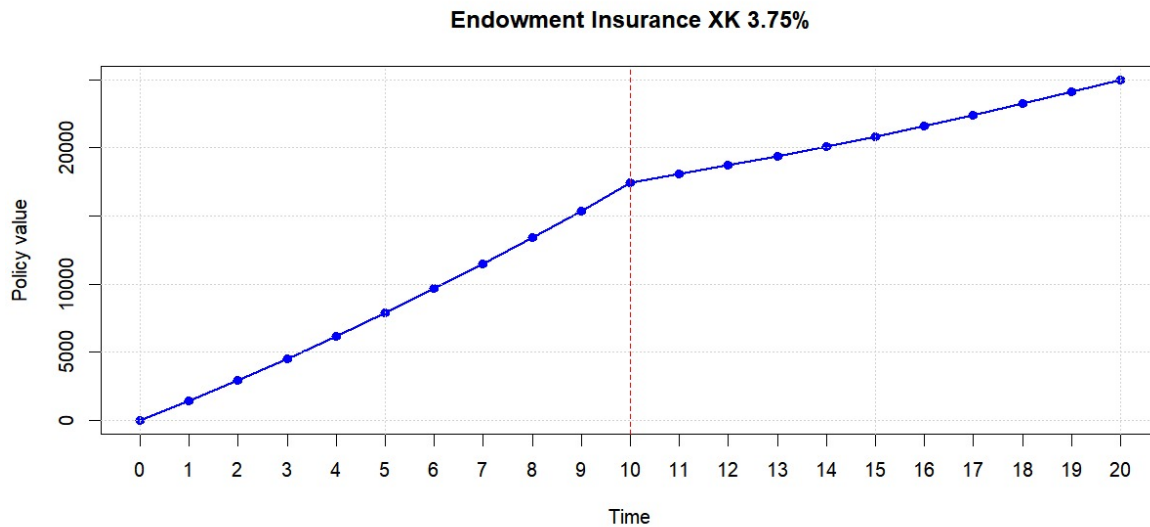


Figure 1: Evolution of Policy Value ($i = 3.75\%$)

We notice that the policy values increase over time. This shows that premiums accumulate over time and reserves are gradually built up. At year 10, the premium-paying period ends, causing a visible change in the slope of the policy value curve, as the reserve growth is no longer driven by incoming premiums but only by interest accumulation and risk of evolution.

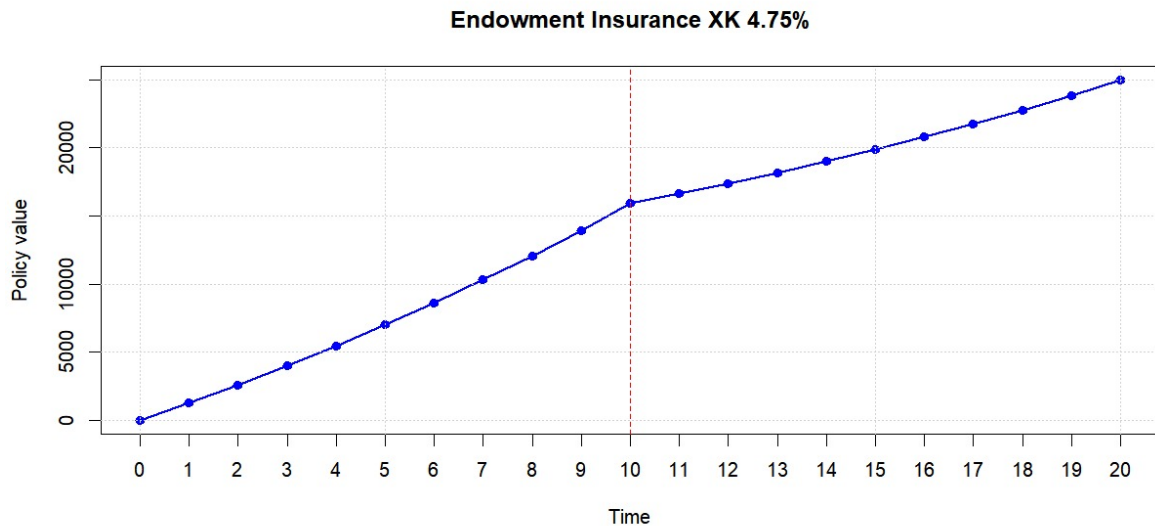
3.1.2 Impact of Increased Interest Rate ($i = 4.75\%$)

We now consider the scenario with an increased interest rate of 4.75%. The calculated policy values are shown in Table 8.

Table 8: Policy Values with increased interest rate ($i = 4.75\%$)

Time (t)	MK Policy Value	FK Policy Value	XK Policy Value
0	0.00	0.00	0.00
1	1,279.47	1,271.03	1,276.52
2	2,619.46	2,602.67	2,613.58
3	4,023.09	3,997.90	4,014.27
4	5,493.71	5,459.88	5,481.87
5	7,034.92	6,991.97	7,019.89
6	8,650.61	8,597.72	8,632.10
7	10,344.98	10,280.94	10,322.56
8	12,122.61	12,045.68	12,095.68
9	13,988.48	13,896.31	13,956.22
10	15,948.05	15,837.48	15,909.35
11	16,665.06	16,568.69	16,631.33
12	17,416.67	17,334.96	17,388.07
13	18,205.05	18,138.20	18,181.65
14	19,032.63	18,980.52	19,014.39
15	19,902.12	19,864.18	19,888.84
16	20,816.57	20,791.69	20,807.86
17	21,779.40	21,765.80	21,774.64
18	22,794.51	22,789.55	22,792.77
19	23,866.35	23,866.35	23,866.35
20	25,000.00	25,000.00	25,000.00

Below is the graph for the higher interest rate $i = 4.75\%$:

Figure 2: Graph for higher interest rate $i = 4.75\%$

We notice that for a higher interest rate we get lower policy values during the term as the graph with $i = 4.75\%$ lies below the base graph with $i = 3.75\%$. This is because

with a higher interest rate, the insurer expects better investment returns. Therefore, less reserve needs to be built today, which lowers interim policy values.

And for only **men** insured (interest rate 3.75%):

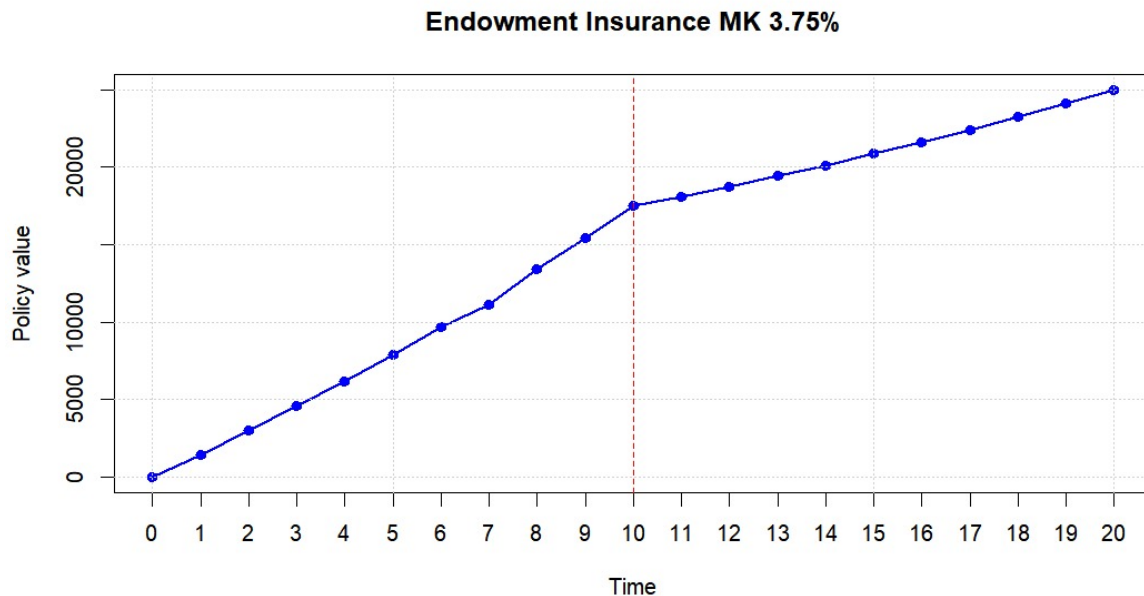


Figure 3: Graph for only men insured (interest rate 3.75%)

The “only men insured” graph lies slightly above the mixed XK graph. Higher male mortality increases the value of the death benefit component. Therefore, more reserve is needed earlier.

3.1.3 Conclusions:

The graphs show that policy values increase steadily over time, are lower under higher interest rates due to stronger discounting, and always converge to the guaranteed capital at maturity, confirming the stability and consistency of the reserving method. The guaranteed sum assured (in our case 25000 EUR) must always be paid at maturity (20 years).

3.2 Profit Sharing

An example of granting profit sharing after the fifth year and calculating the impact on the insured guarantees.

We have:

- Dotation $d = 2\%$
- ${}_5V = 7906.42$

- ${}_{15}E_{40}(XK_{i^*=3.65}) = 0.5329968$
- $A_{40:\overline{15}|}^1(XK_{i^*=3.65}) = 0.05304$

By the formula:

$$\Delta KL = \frac{d}{{}_{n-t}E_{x+t} + A_{x+t:n-t}^1}$$

We get the outcome:

$$\begin{aligned} \Delta KL &= \frac{d}{{}_{15}E_{40} + A_{40:\overline{15}|}^1} \\ &= \frac{2\%}{0.5329968 + 0.05304} \\ &\approx 269.83 EUR \end{aligned}$$

3.3 Guarantee System

Indicate the type of guaranteed system of the insurance:

We are using the **classic system**.

Our contract uses a fixed premium plan where one premium covers both survival benefit and death benefit, which is a key feature of the classic system. The product promises a guaranteed life capital at maturity and a guaranteed death capital in case of death before maturity. This is fully aligned with the classic system where benefits are guaranteed in advance. Furthermore, the guarantees apply to both paid and future premiums. The classic system is associated with low but stable performance, which fits our product's objective of providing security and capital protection through the Branch 21 component. In our product design, the technical interest rate and internal charges are not explicitly communicated to the policyholder, and the price of death coverage is not separated, which matches the classic system's approach. Our product communicates mainly through annual information on surrender values and profit-sharing, which corresponds to the classic system framework.

3.4 Use Case Scenarios

Here are other examples when this insurance can be a suitable solution:

Parents planning children's education: Parents can purchase this 20-year endowment policy when their child is born, setting the maturity date to coincide with the child's entry into university. This ensures a guaranteed lump sum to fund education expenses at maturity. At the same time, the policy provides life protection during the parents' working years, so if a parent passes away before maturity, the family still receives financial support.

Individuals saving for retirement supplements: Individuals in their 40s or 50s can take out a 20-year endowment policy designed to mature around age 60 or 65. This structure provides both life insurance coverage throughout the policy term and a guaranteed lump sum at maturity, offering a reliable supplement to retirement income. It combines protection with disciplined savings, ensuring financial security in later life.

3.5 Key Business Challenges

Two important challenges for your insurance company are:

1. Mortality Risk: If actual death rates turn out higher than expected, the insurer must pay the sum assured earlier than anticipated. This shortens the time available to invest and grow the collected premiums, reducing the opportunity for investment income. The resulting increase in claims directly erodes profitability and puts pressure on the company's reserves.

2. Investment / Interest Rate Risk: Endowment policies promise a guaranteed lump sum at maturity. If investment returns fall short of the assumed interest rate, the insurer may struggle to generate sufficient funds to meet these guarantees. Prolonged low-interest environments are particularly challenging, as assets may not yield enough to cover the guaranteed benefits, forcing the insurer to absorb the shortfall and weakening financial performance.